SQL Assignment and Journal

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-- Mine Code

-- 1 . Transform

SELECT customerName,
CASE
    WHEN country = 'USA' THEN 'Local'
    ELSE 'International'
END AS customer_type
FROM customers;

-- order status cleaning
SELECT orderNumber, status,
CASE
    WHEN status = 'Shipped' THEN 'Completed'
    ELSE 'Pending'
END AS order_status_clean
FROM orders;

-- 2 . Filtering

-- Only shipped order
SELECT *
FROM orders
WHERE status = 'Shipped';

-- 3. Joining gareko
-- Joining Customer and Orders Table
SELECT c.customerName, o.orderNumber, o.status
FROM customers c
JOIN orders o
ON c.customerNumber = o.customerNumber;

-- Customers & Payments Table join gareko
SELECT c.customerName, p.amount
FROM customers c
JOIN payments p
ON c.customerNumber = p.customerNumber;

-- Customer and OrderDetails join gareko
SELECT o.orderNumber, od.productCode, od.quantityOrdered
FROM orders o
JOIN orderdetails od
ON o.orderNumber = od.orderNumber;

-- 4 . Aggregate

SELECT SUM(amount) AS total_sales
FROM payments;

SELECT customerNumber, COUNT(orderNumber) AS total_orders
FROM orders
GROUP BY customerNumber;

SELECT customerNumber, SUM(amount) AS total_payment
FROM payments
GROUP BY customerNumber;

SELECT salesRepEmployeeNumber, COUNT(customerNumber) AS total_customers
FROM customers
GROUP BY salesRepEmployeeNumber;

SELECT SUM(quantityOrdered * priceEach) AS total_sales
FROM orderdetails;
```

SQL Practice Journal In SQL, I worked with the Classic Models database. I learned how to transform, join, and aggregate data. I used CASE statements to clean and categorize data, such as grouping customers into Local and International. I used filtering to select specific records like shipped orders. I practiced JOIN operations to connect multiple tables such as customers, orders, and payments. This helped me understand relationships between data. I also used aggregate

functions like SUM and COUNT to calculate total sales and number of orders. I learned how to use GROUP BY to summarize data. Challenges Faced I faced some confusion while understanding JOIN conditions and how tables are connected. I also made small mistakes in column names and syntax, but I fixed them by checking errors and correcting them step by step.